

VisionSentry

Thermal & RGB UAV Detection, Tracking and Distance Estimation

Endrit Shaqiri · James Njoroge · Muhammad Raka Zuhdi
Mofolaoluwarera Oladipo · Jitarth Patel

Boston University · CS 585 · Spring 2026

Problem & Motivation

- Small UAVs occupy only a tiny region of each frame making them hard to detect reliably
- RGB cameras degrade in low-light and night conditions; IR cameras maintain visibility but lose spatial detail
- Most detection systems handle a single modality with no range awareness
- Goal: build a complete, modular UAV perception system across both IR and RGB



Drone visible in IR (right), obscured in RGB (left)

Small Object Detection

UAVs often < 20px in frame

Low-Light Failure

RGB unusable at night

No Range Awareness

2D boxes give no depth

What We Built

Anti-UAV
Dataset

YOLO
Conversion

IR Detector
YOLO v12n

RGB Detector
YOLO 11x

BoT-SORT
Tracking

Distance
Estimation

Gradio
Web App

IR Branch

- Anti-UAV dataset (318k IR frames)
- YOLO v12n
- $mAP@0.50 = 0.919$

RGB Branch

- Anti-UAV-RGBT dataset
- YOLO 11x
- Heatmap visualization

Shared: BoT-SORT tracker · Distance estimation module · Gradio web app

Datasets

Anti-UAV — Thermal/IR

- 318,932 training images · 80,478 val images
- 189 train sequences · 50 val sequences
- Single class: uav — 96% of frames labeled
- 3 raw annotation formats unified into one YOLO layout
- Sequence-level split — no leakage from adjacent frames

Anti-UAV — RGB+ Thermal/IR

- Paired RGB + thermal UAV sequences
- Benchmark from emergentmind.com
- Same YOLO label format as IR branch
- Enables direct IR vs. RGB comparison
- Same sequence-level split strategy applied



Why sequence-level split?

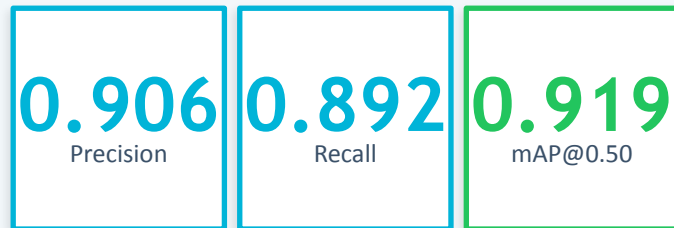
Adjacent frames in a video are nearly identical so a frame-level split would leak the training set into validation, inflating metrics. Splitting by sequence keeps entire videos in one split.

Detection: YOLO v12n (IR) & YOLO 11x (RGB)

Architecture — YOLO family

- CSP backbone extracts feature maps at multiple scales; coarse for context, fine for small targets
- Neck (FPN/PAN) fuses those scales together so tiny UAVs aren't lost in downsampling
- Decoupled head handles box location and class label separately — more accurate than doing both at once
- Anchor-free — predicts the drone center directly instead of adjusting pre-set anchor boxes
- Loss combines CloU (box shape), BCE (classification), and DFL (box edge sharpness)

IR Model Results



2-epoch smoke test at 640px

IR — YOLO v12n

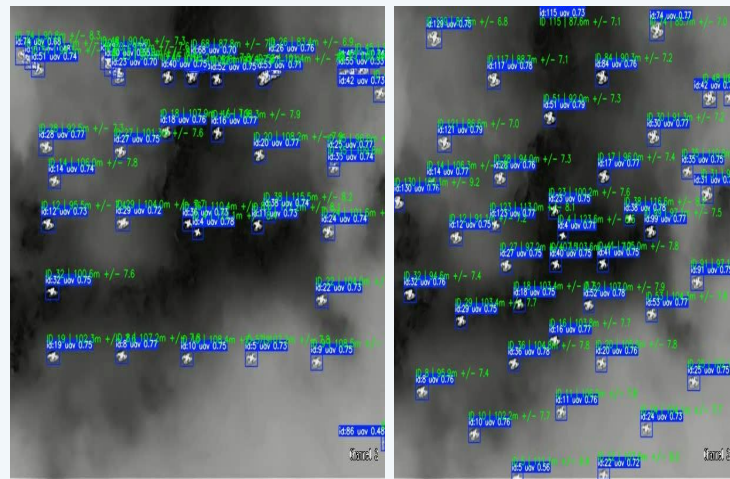
- Pretrained COCO → fine-tuned Anti-UAV
- 960px · batch 16 · 100 epochs · patience 30
- AdamW · lr 0.01 · BU SCC (SLURM)

RGB — YOLO 11x

- Pretrained weights → Anti-UAV-RGBT
- 640px · batch 8 · 32 epochs
- AdamW · lr 0.001 · BU SCC

Tracking: BoT-SORT

- BoT-SORT is an online multi-object tracker: runs frame-by-frame with no future knowledge
- Kalman filter maintains a state estimate (position + velocity) per track between detections
- Hungarian algorithm matches new detections to existing tracks via IoU cost matrix
- track_buffer: track kept alive for **60** frames without a detection (handles occlusion)
- gmc_method: (None) global motion compensation via sparse optical flow is unreliable on low-texture IR
- Exports MOT format: frame, id, x, y, w, h, score per row



Real tracking output — IR frames with persistent track IDs + distance

1,035

Frames Processed

629

MOT Rows

56

Unique Track IDs

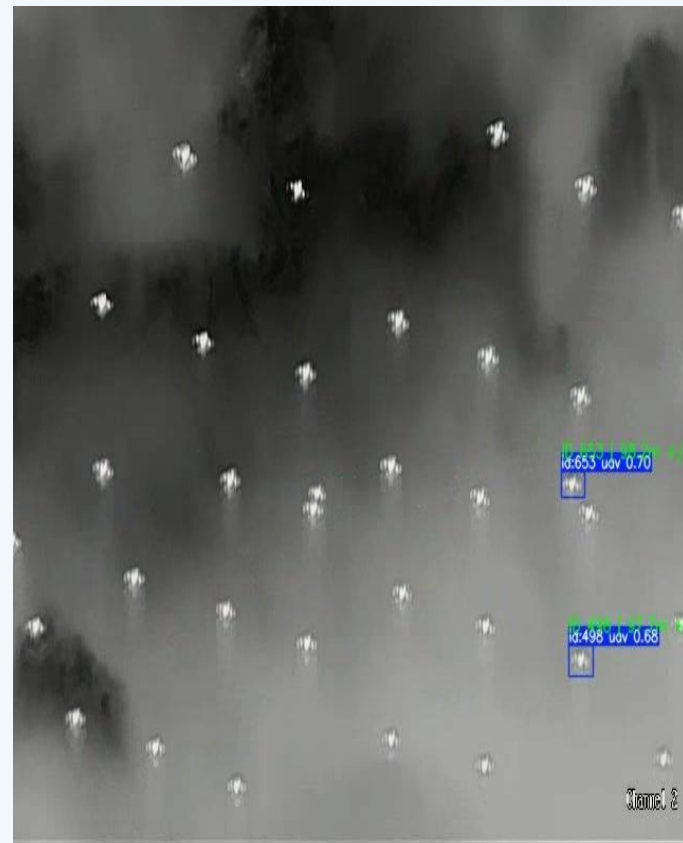
Distance Estimation

Monocular Geometric Estimation

$$d = (f \times W_{\text{real}}) / w_{\text{px}}$$

f = focal length (px) · W_{real} = known UAV width (m) · w_{px} = bbox width in pixels

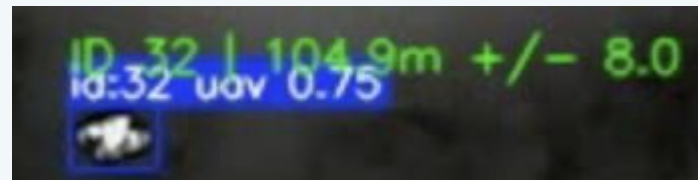
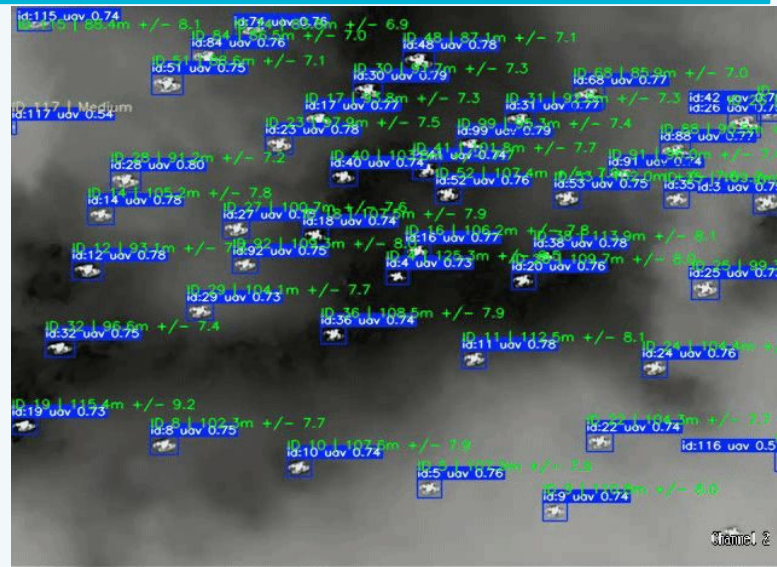
- If a drone looks tiny in frame, it's far so the module uses known drone size + bbox width to estimate range
- Kalman filter smooths the distance per track so the number doesn't jump frame to frame
- When the box is too small to trust, falls back to "close / medium / far" instead of a bad metric
- No camera calibration needed, focal length is estimated from field of view, with added uncertainty



Live: ID · distance · $\pm$ uncertainty per track

System in Action

- Real Anti-UAV thermal footage with 1,035 frames processed
- visionsentry-deployment.onrender.com
- [Video demo](#)
- Blue label (drawn by YOLO at the top-left of each box):
 - uav: detected class.
 - 0.75: detector confidence (range 0 to 1).
 - id:32: persistent track ID assigned by BoT-SORT (same drone keeps the same ID across frames).
- Colored text above the blue label (only when Include Distance Estimation is on): ID 32 | 12.5m +/- 2.1
 - ID 32: Persistent Track ID
 - 104.9m: estimated distance to the drone in meters.
 - +/- 8.0: Standard deviation of the distance estimate in meters. Smaller is more confident.



What We Achieved

- End-to-end UAV perception system from raw dataset to live web app
- 318,932 IR training frames converted, cleaned, and sequence-split
- YOLO v12n IR detector: mean Average Precision@0.50 = 0.919, Precision = 0.906, Recall = 0.892
- BoT-SORT tracker with Kalman filtering across 1,035 frames
- Monocular distance estimation with uncertainty running live in the web app